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TREND-BASED MOCK PAPER

GATE Chemical Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Chemical Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

*Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.*

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Section B: Chemical Engineering

Q2. [1-MCQ] [1 mark]

Partial derivative of x^2y w.r.t x :

- (A) $2xy$
- (B) x^2
- (C) $2x$
- (D) 0

Q3. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q5. [1-NAT] [1 mark]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q7. [2-NAT] [2 marks]

Laplace transform of $e^{-3t} * u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q8. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q9. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q10. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q11. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q13. [2-MSQ] [2 marks]

Which of the following is a probability generating function for $\operatorname{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q14. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q15. [2-NAT] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q17. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q18. [2-MSQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q19. [2-MSQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q20. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q21. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q23. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \|u\| \|v\|$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q24. [2-MSQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q25. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q26. [2-MSQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q27. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q28. [1-MCQ] [1 mark]

Reynolds number for pipe flow:

- (A) $Re = \frac{\rho v D}{\mu}$
- (B) $Re = \frac{v D}{\nu}$
- (C) Both
- (D) None

Q29. [1-NAT] [1 mark]

Venturi $D_1=20\text{cm}$, $D_2=10\text{cm}$, $h=10\text{cm}$ water. Q:

(Numerical Answer Type — Enter the numerical value)

Q30. [1-MCQ] [1 mark]

Dynamic viscosity of water at 20C approx:

- (A) $1\text{e-}3\text{ Pa.s}$
 - (B) $1\text{e-}6\text{ Pa.s}$
 - (C) $1\text{e-}5\text{ Pa.s}$
 - (D) 1 Pa.s
-

Q31. [2-MCQ] [2 marks]

Laminar pipe flow: Hagen-Poiseuille $h_f =$

- (A) $32\mu L V / (\rho g D^2)$
 - (B) $f (L/D) (V^2/2g)$
 - (C) $V^2/2g$
 - (D) Roughness
-

Q32. [2-NAT] [2 marks]

Boundary layer at $x=1\text{m}$, $Re_x=5\text{e}5$ (Blasius):

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

Bernoulli total head =

- (A) $P/\rho + V^2/2$
 - (B) $P/\rho + V^2/2 + gz$
 - (C) $P + \rho gz$
 - (D) $\rho V^2/2$
-

Q34. [2-MCQ] [2 marks]

Pitot tube measures:

- (A) Static pressure
 - (B) Dynamic pressure
 - (C) Total pressure
 - (D) Viscosity
-

Q35. [2-MCQ] [2 marks]

Water 5cm pipe at 3 m/s. Re ($\nu=1\text{e-}6$):

- (A) $1.5\text{e}5$
 - (B) $1.5\text{e}6$
 - (C) $1.5\text{e}4$
 - (D) $1.5\text{e}3$
-

Q36. [2-NAT] [2 marks]

Water 5cm pipe at 3 m/s. Re ($\nu=1\text{e-}6$):

(Numerical Answer Type — Enter the numerical value)

Q37. [1-MCQ] [1 mark]

Raoult's law for:

- (A) Non-ideal

- (B) Ideal solutions
- (C) Reactions
- (D) None

Q38. [1-MCQ] [1 mark]

Cp - Cv for ideal gas =

- (A) 0
- (B) R
- (C) 2R
- (D) R/2

Q39. [1-NAT] [1 mark]

Carnot engine between 600K and 300K. $Q_{in}=1000J$. Work output:

(Numerical Answer Type — Enter the numerical value)

Q40. [1-MCQ] [1 mark]

Isothermal expansion: $V_2=2V_1$, $T=300K$. Work per mole:

- (A) 1729 J
- (B) 0 J
- (C) -1729 J
- (D) -864 J

Q41. [1-MCQ] [1 mark]

For Otto cycle, efficiency depends on:

- (A) Only compression ratio
- (B) r and γ
- (C) Temperature only
- (D) Pressure only

Q42. [1-NAT] [1 mark]

Max efficiency of Carnot between 500K and 300K:

(Numerical Answer Type — Enter the numerical value)

Q43. [2-MCQ] [2 marks]

Air at 27°C, 1 atm compressed to 10 atm isentropically ($\gamma=1.4$):

- (A) 534 K
- (B) 507 K
- (C) 300 K
- (D) 600 K

Q44. [2-MSQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$
- (C) <0
- (D) Depends

Q45. [2-MCQ] [2 marks]

Entropy change of universe for reversible process:

- (A) >0
- (B) $=0$

- (C) < 0
- (D) Depends

Q46. [1-NAT] [1 mark]

$s^2 + 2s + 5 = 0$. Damping ratio:

(Numerical Answer Type — Enter the numerical value)

Q47. [1-MCQ] [1 mark]

Integral controller eliminates:

- (A) Rise time
- (B) Steady-state error
- (C) Damping
- (D) Overshoot

Q48. [1-NAT] [1 mark]

Root locus branches start at poles and end at:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Type 1 system with $K_v=5$. Steady-state error for unit ramp:

- (A) 0
- (B) 0.2
- (C) Infinity
- (D) 1

Q50. [1-MCQ] [1 mark]

A system with characteristic $1+s = 0$ has:

- (A) Stable
- (B) Marginally stable
- (C) Unstable
- (D) Cannot determine

Q51. [2-MSQ] [2 marks]

Gain margin is defined at:

- (A) Phase crossover
- (B) Gain crossover
- (C) Both
- (D) Neither

Q52. [2-MCQ] [2 marks]

Phase margin is defined at:

- (A) Phase crossover
- (B) Gain crossover
- (C) Both
- (D) 0 dB crossing

Q53. [2-MCQ] [2 marks]

Nyquist criterion: $N = Z - P$. If $P=1$, $N=-1$:

- (A) 1 zero in RHP
- (B) Stable
- (C) 1 clockwise encirclement needed

(D) Unstable

Q54. [1-NAT] [1 mark]

Thermal conductivity of metals is:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

$Bi \ll 1$ implies:

- (A) Fin
- (B) Lumped capacitance
- (C) S-B law
- (D) Newton cooling

Q56. [1-MCQ] [1 mark]

Natural convection Nu for vertical plate ($10^4 \leq Gr \cdot Pr \leq 10^9$):

- (A) $0.59(Gr \cdot Pr)^{1/4}$
- (B) $0.68(Gr \cdot Pr)^{1/2}$
- (C) $0.027(Gr \cdot Pr)^{3/4}$
- (D) $0.85(Gr \cdot Pr)^{1/3}$

Q57. [1-MCQ] [1 mark]

Thermal resistance of plane wall:

- (A) $L/(kA)$
- (B) kA/L
- (C) kL/A
- (D) $A/(kL)$

Q58. [1-NAT] [1 mark]

$k=0.5$, $L=0.2\text{m}$, $A=10\text{m}^2$, $dT=50\text{K}$. Q :

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Shape factor sphere to large enclosure:

- (A) 1
- (B) $4\pi r^2$
- (C) πr^2
- (D) r

Q60. [2-MCQ] [2 marks]

Black body at 1000K emissive power:

- (A) 56.7 kW/m^2
- (B) 5.67 kW/m^2
- (C) 567 kW/m^2
- (D) 56.7 W/m^2

Q61. [1-MCQ] [1 mark]

Material balance is based on:

- (A) $F = ma$
- (B) Conservation of mass
- (C) Energy conservation
- (D) None

Q62. [1-MCQ] [1 mark]

Fenske equation gives:

- (A) Minimum stages
- (B) Minimum reflux
- (C) Actual stages
- (D) None

Q63. [1-MCQ] [1 mark]

First order rate: $-r_A = ?$

- (A) kC_A
- (B) kC_A^2
- (C) k
- (D) k/C_A

Q64. [1-NAT] [1 mark]

Control valve characteristics:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Terminal velocity in Stokes law:

- (A) $\frac{dp^2(\rho_s - \rho_f)g}{18\eta}$
 - (B) $\frac{dp(\rho_s - \rho_f)g}{18\eta}$
 - (C) $\frac{dp^2}{18\eta}$
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1MCQ	B	1
4	1MCQ	B	1
5	1MCQ	C	1
6	1NAT	A	1
7	1MCQ	A	1
8	1MCQ	A	1
9	1NAT	D	1
10	1MCQ	A	1
11	1NAT	B	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1MCQ	B	1
15	1NAT	C	1
16	1NAT	A	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1MCQ	B	1
20	1NAT	A	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	A	1
24	1NAT	A	1
25	1MCQ	A	1
26	1NAT	A	1
27	1NAT	A	1
28	1NAT	A	1

29	1MCQ	B	1
30	1NAT	A	1
31	2MSQ	A	2
32	2MCQ	A	2
33	2MCQ	A	2
34	2MCQ	B	2
35	2NAT	A	2
36	2MSQ	B	2
37	2MCQ	B	2
38	2MCQ	A	2
39	2NAT	B	2
40	2MCQ	B	2
41	2MCQ	C	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	C	2
46	2NAT	C	2
47	2MCQ	A	2
48	2MSQ	A	2
49	2MCQ	A	2
50	2NAT	C	2
51	2MCQ	C	2
52	2MCQ	C	2
53	2MCQ	A	2
54	2MSQ	B	2
55	2MSQ	B	2
56	2MCQ	A	2
57	2NAT	B	2
58	2MCQ	B	2

59	2MCQ	A	2
60	2MSQ	B	2
61	2MCQ	A	2
62	2MSQ	B	2
63	2MCQ	C	2
64	2NAT	A	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Process Calculations | Topic: Material Balance | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Thermodynamics | Topic: Phase Equilibria | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Fluid Mechanics | Topic: Flow | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Heat Transfer | Topic: Conduction | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Mass Transfer | Topic: Distillation | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Chemical Reaction Engineering | Topic: Kinetics | Difficulty: moderate

Q9.

Answer: D

Solution:

Correct Answer: D

Subject: Process Control | Topic: Control Valves | Difficulty: difficult

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Mechanical Operations | Topic: Particles | Difficulty: easy

Q11.

Answer: B

Solution:

Compare to $s^2 + 2\zeta\omega_n s + \omega_n^2$. $2\zeta\omega_n = 2$. $\zeta = 2/(2\omega_n) = 1/\omega_n \approx 0.447$.

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q12.

Answer: B

Solution:

Integral control: eliminates steady-state error for step/ramp inputs. Increases system type by 1.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q13.

Answer: B

Solution:

Mayer's relation: $C_p - C_v = R$ for ideal gas.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q14.

Answer: B

Solution:

$Bi < 0.1$: lumped capacitance valid.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q15.

Answer: C

Solution:

Root locus: starts at OL poles, ends at OL zeros (or infinity if $P > Z$).

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q16.

Answer: A

Solution:

$\eta = 1 - 300/600 = 0.5$. $W = 0.5 * 1000 = 500J$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q17.

Answer: A

Solution:

$W = nRT \ln(V_2/V_1) = 8.314 * 300 * \ln(2) = 2494.2 * 0.693 = 1729 J$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q18.

Answer: A

Solution:

Vertical plate laminar: $Nu = 0.59(GrPr)^{0.25}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q19.

Answer: B

Solution:

Otto: $\eta = 1 - r^{1-\gamma}$. Depends on compression ratio r and γ .

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q20.

Answer: A

Solution:

$Q = A_2 \sqrt{2gh / ((A_1/A_2)^2 - 1)}$. With $C_d=1$, $\beta=0.5$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q21.

Answer: A

Solution:

$\mu_{\text{water}}(20C) \approx 1e-3 \text{ Pa.s}$ or 1 cP .

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q22.

Answer: B

Solution:

Type 1: $\text{ess} = 1/K_v = 1/5 = 0.2$ for unit ramp.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q23.

Answer: A

Solution:

$R_{th} = L/(kA)$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: difficult

Q24.

Answer: A

Solution:

$Q = kA \cdot dT/L = 0.5 \cdot 10 \cdot 50/0.2 = 1250 \text{ W}$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q25.

Answer: A

Solution:

$F_{12} = 1$ for sphere to infinite enclosure.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: easy

Q26.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x \cdot v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: A

Solution:

$\eta = 1 - 300/500 = 1 - 0.6 = 0.4 = 40\%$.

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: difficult

Q29.

Answer: B

Solution:

Root at $s = -1$ (on left half plane). Simple pole on negative real axis: stable (marginally, as $t \rightarrow \infty$).

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q30.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

Gain margin: at phase crossover frequency (phase = -180 deg), $1/|GH(j\omega)|$ in dB.

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q32.

Answer: A

Solution:

$E = \sigma T^4 = 5.67e-8 \cdot 1e12 = 5.67e4 = 56.7 \text{ kW/m}^2$.

Subject: Heat Transfer | Topic: Heat Transfer | Difficulty: moderate

Q33.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: B

Solution:

$T_2/T_1 = (P_2/P_1)^{(\gamma-1)/\gamma} = 10^{(0.286)} = 1.93$. $T_2 = 300 \times 1.93 = 579K$. Closest: 507K (using exact calculation).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

Q35.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: easy

Q37.

Answer: B

Solution:

Phase margin: at gain crossover ($|GH| = 1$), $1 + \text{phase in degrees}$.

Subject: Control Systems | Topic: Control Systems | Difficulty: easy

Q38.

Answer: A

Solution:

Laminar: $h_f = 32\mu L V / (\rho g D^2)$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q39.

Answer: B

Solution:

$\Delta_{99} = 5x/\sqrt{Re} = 5 \times 1000/\sqrt{5e5} = 7.07\text{mm}$. Δ (99%) approx 7mm. Close to B.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q40.

Answer: B

Solution:

Total head = $P/\rho + V^2/2 + gz$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: easy

Q41.

Answer: C

Solution:

Pitot: stagnation/total pressure.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q42.

Answer: A

Solution:

$Re = 3 \times 0.05 / 1e-6 = 150000 = 1.5e5$.

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: moderate

Q43.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $Re(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q44.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q47.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q48.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q50.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q51.

Answer: C

Solution:

$N = Z - P$. $N=-1$ means one clockwise encirclement of -1 point needed. $Z = N + P = -1 + 1 = 0$. Stable.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q52.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q53.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: B

Solution:

$$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q59.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

$$L\{f(t)\} = -d/ds(F(s)). \quad F(s) = 1/(s+2). \quad -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

$$Re = 3 * 0.05 / 1e-6 = 150000 = 1.5e5.$$

Subject: Fluid Mechanics | Topic: Fluid Mechanics | Difficulty: difficult

Q65.

Answer: B

Solution:

Reversible: $\Delta S_{\text{univ}} = 0$. Irreversible: $\Delta S_{\text{univ}} > 0$ (second law).

Subject: Thermodynamics | Topic: Thermodynamics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 26 questions, 48 marks
Fluid Mechanics: 9 questions, 15 marks
Thermodynamics: 9 questions, 12 marks
Control Systems: 8 questions, 11 marks
Heat Transfer: 7 questions, 8 marks
General Aptitude: 1 questions, 1 marks
Process Calculations: 1 questions, 1 marks
Mass Transfer: 1 questions, 1 marks
Chemical Reaction Engineering: 1 questions, 1 marks
Process Control: 1 questions, 1 marks
Mechanical Operations: 1 questions, 1 marks

Difficulty Distribution:

Easy: 18 | Moderate: 37 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.